



US012401934B2

(12) **United States Patent**
Huang et al.

(10) **Patent No.:** **US 12,401,934 B2**

(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **SEMICONDUCTOR DEVICE PACKAGE AND ACOUSTIC DEVICE HAVING THE SAME**

(71) Applicant: **Advanced Semiconductor Engineering, Inc.**, Kaohsiung (TW)

(72) Inventors: **Ming-Tau Huang**, Kaohsiung (TW);
Kuei-Hao Tseng, Kaohsiung (TW);
Hung-I Lin, Kaohsiung (TW)

(73) Assignee: **ADVANCED SEMICONDUCTOR ENGINEERING, INC.**, Kaohsiung (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 176 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **17/968,725**

(22) Filed: **Oct. 18, 2022**

(65) **Prior Publication Data**

US 2023/0039552 A1 Feb. 9, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/528,338, filed on Jul. 31, 2019, now Pat. No. 11,477,559.

(51) **Int. Cl.**
H04R 1/10 (2006.01)
B81B 7/00 (2006.01)

(52) **U.S. Cl.**
CPC **H04R 1/1016** (2013.01); **B81B 7/008** (2013.01); **H04R 2201/003** (2013.01)

(58) **Field of Classification Search**
CPC .. H04R 1/1016; H04R 2201/003; B81B 7/008

USPC 381/74, 58, 151, 311, 380, 374, 150;
438/106, 107, 112; 343/791, 866;
455/572; 320/127, 128, 137

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,570,218 B2 8/2009 Tsujimura et al.
8,315,060 B2 11/2012 Morikita et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN 107623881 A 1/2018
CN 208754499 U 4/2019

(Continued)

OTHER PUBLICATIONS

First Office Action, with English summary for counterpart Japanese Patent Application No. 2019-198831, issued Oct. 31, 2023, 9 pages.

(Continued)

Primary Examiner — Vivian C Chin

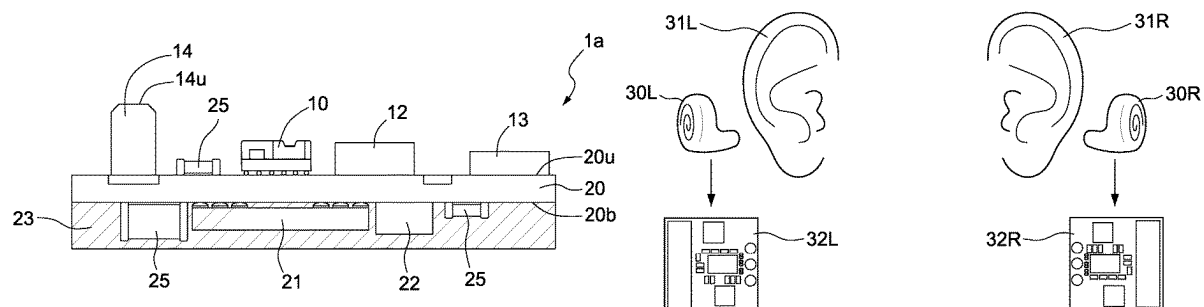
Assistant Examiner — Con P Tran

(74) *Attorney, Agent, or Firm* — FOLEY & LARDNER LLP

(57) **ABSTRACT**

A semiconductor device package comprises a first substrate, a first electronic component, an encapsulant, and a feeding structure. The first substrate has a first surface and a second surface opposite the first surface. The first electronic component is disposed on the first surface of the first substrate. The encapsulant encapsulates the first electronic component on the first surface of the first substrate. The feeding structure is disposed on the second surface of the first substrate without covering.

20 Claims, 10 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

9,118,990	B2	8/2015	Hankey et al.
9,932,221	B1	4/2018	Natan et al.
10,049,893	B2	8/2018	Chen et al.
11,477,559	B2	10/2022	Huang et al.
2003/0169209	A1	9/2003	Ohara et al.
2004/0084771	A1	5/2004	Bolken et al.
2007/0029667	A1	2/2007	Fujii et al.
2007/0080360	A1	4/2007	Mirsky et al.
2008/0164545	A1	7/2008	Hsiao
2012/0049348	A1	3/2012	Ro
2014/0198927	A1	7/2014	Cheng
2015/0350762	A1	12/2015	Birger et al.
2016/0050474	A1	2/2016	Rye et al.
2016/0073189	A1	3/2016	Linden et al.
2016/0225745	A1	8/2016	Beer et al.
2017/0195767	A1	7/2017	Kim et al.
2017/0209095	A1	7/2017	Wagner et al.
2017/0330870	A1	11/2017	Chen et al.
2018/0323170	A1	11/2018	Kim et al.
2019/0045294	A1	2/2019	Kim et al.
2020/0221210	A1 *	7/2020	Grebe H01Q 1/40

FOREIGN PATENT DOCUMENTS

JP	2001-217332	A	8/2001
JP	2001-352212	A	12/2001
JP	2007-042978	A	2/2007
JP	2007-288360	A	11/2007
JP	2010-516096	A	5/2010
JP	2011-097306	A	5/2011
JP	2012-074505	A	4/2012
JP	2017-204635	A	11/2017
JP	2021-027308	A	2/2021

WO	WO-2005/067359	A1	7/2005
WO	WO-2009/122835	A1	10/2009
WO	WO-2015/068403	A1	5/2015

OTHER PUBLICATIONS

Extended Search Report for corresponding European Patent Application No. 19204047.5, issued on Jun. 5, 2020, 7 pages.

Final Office Action U.S. Appl. No. 16/528,338, issued Nov. 10, 2020, 39 pages.

Final Office Action U.S. Appl. No. 16/528,338, issued, Sep. 14, 2021, 46 pages.

Non-Final Office Action for U.S. Appl. No. 16/528,338, issued May 29, 2020, 20 pages.

Non-Final Office Action U.S. Appl. No. 16/528,338, issued Feb. 3, 2022, 16 pages.

Non-Final Office Action U.S. Appl. No. 16/528,338, issued Mar. 22, 2021, 37 pages.

Notice of Allowance U.S. Appl. No. 16/528,338, issued Jun. 13, 2022, 9 pages.

Office Action for corresponding European Patent Application No. 19204047.5, issued on Jun. 21, 2021, 6 pages.

Office Action, for counterpart Chinese Patent Application No. 201910764286.3, issued Nov. 28, 2024, 7 pages.

Search Report with English translation, counterpart Chinese Patent Application No. 201910764286.3, issued Nov. 28, 2024, 4 pages.

Office Action, for counterpart Japanese Patent Application No. 2024-063839, issued Mar. 5, 2025, 5 pages.

Summary with English Translation of Office Action, for counterpart Japanese Patent Application No. 2024-063839, issued Mar. 5, 2025, 1 page.

Office Action for counterpart European Patent Application No. 19204047.5, issued Jan. 24, 2025, 8 pages.

Search Report with English translation for Chinese Application No. 201910764286.3 dated Jul. 7, 2025, 4 pages.

* cited by examiner

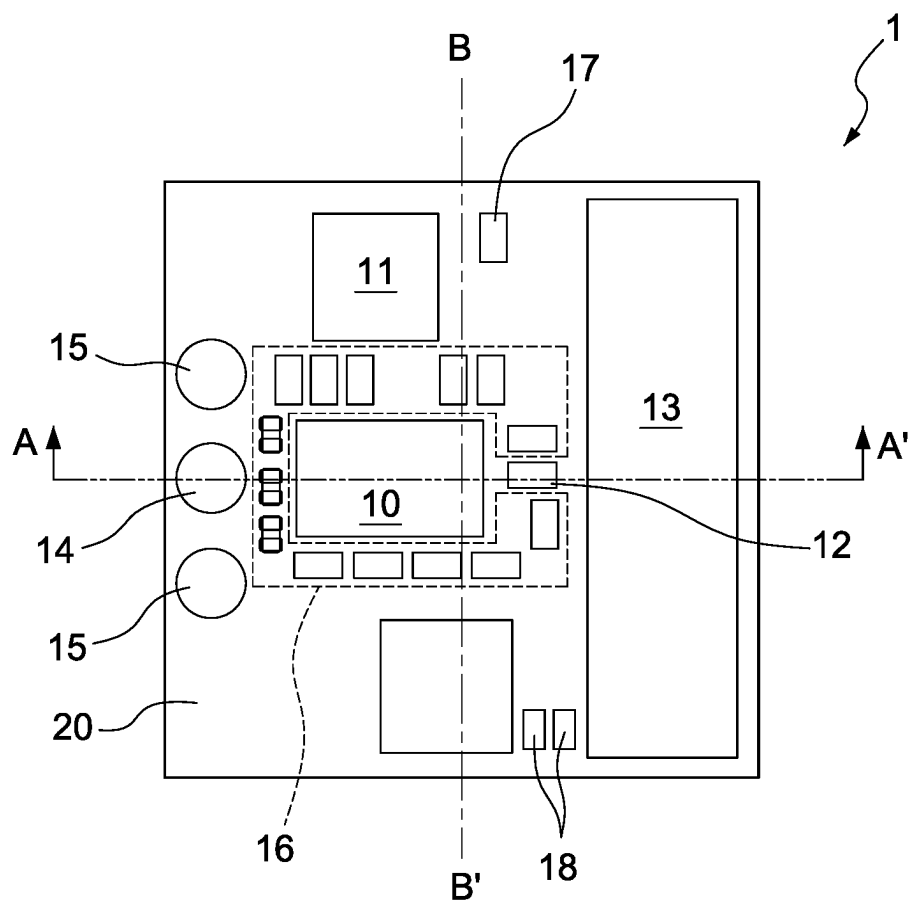


FIG. 1

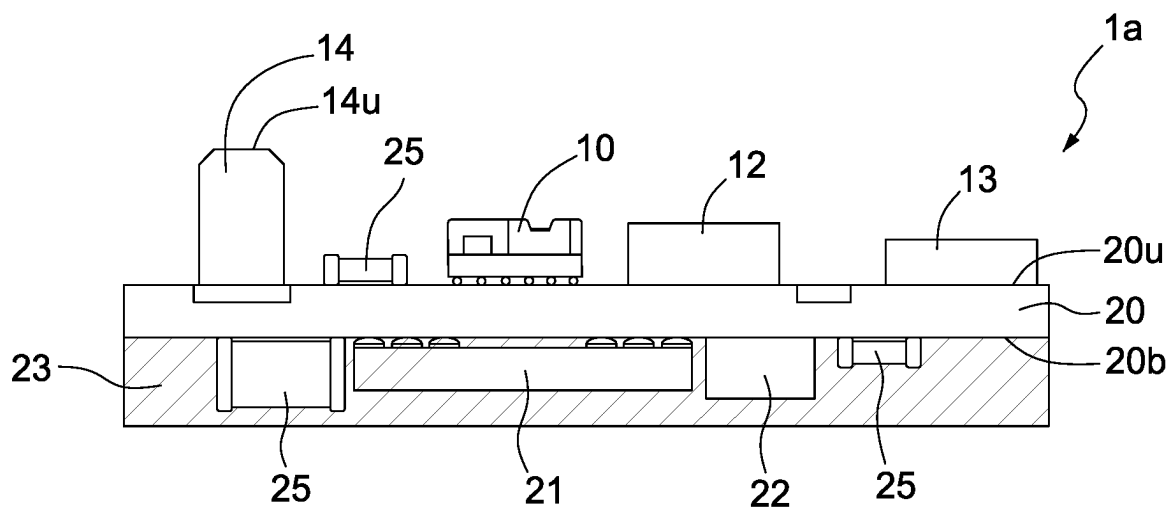


FIG. 2A

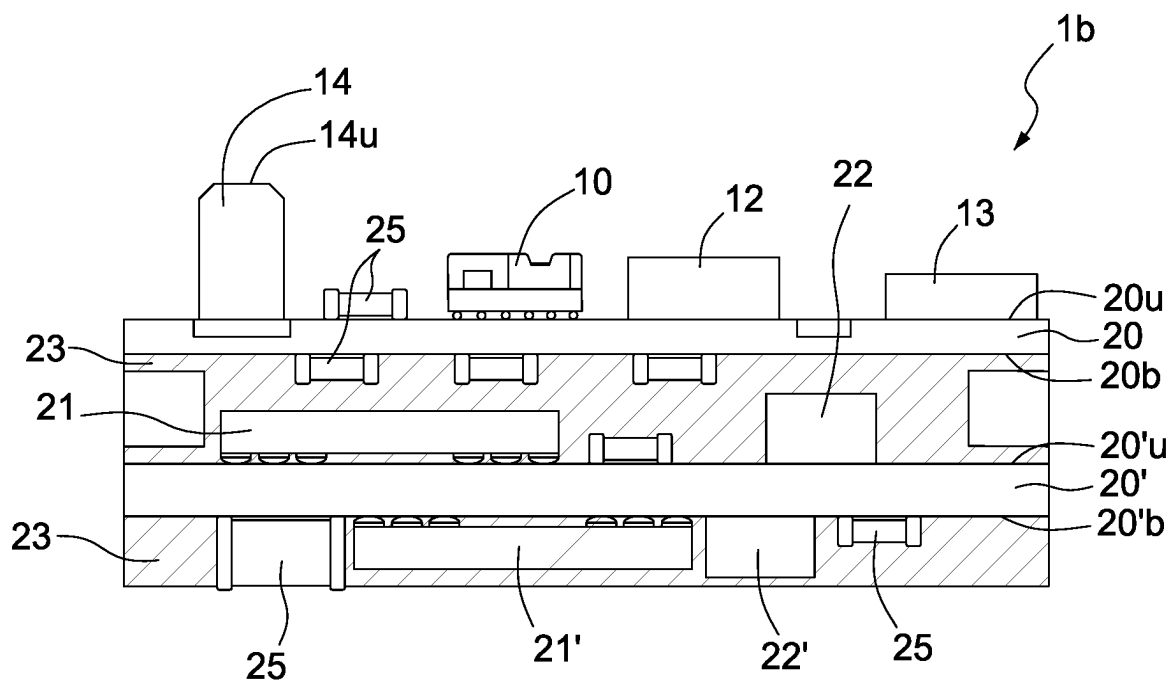


FIG. 2B

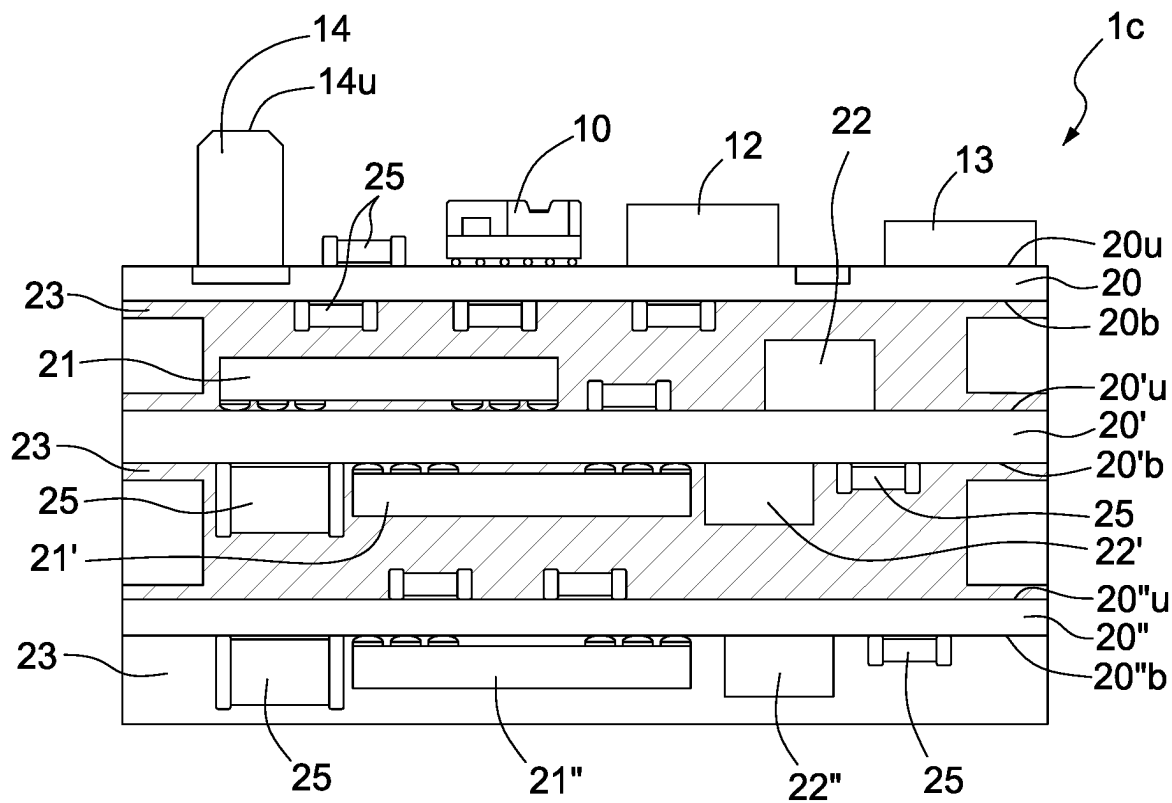


FIG. 2C

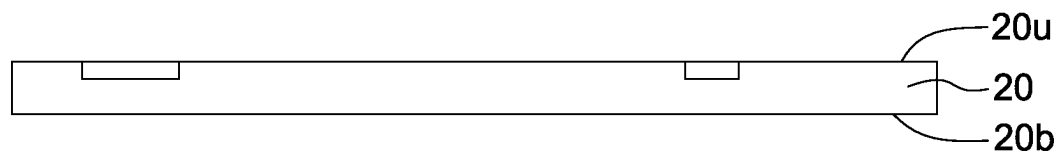


FIG. 2D

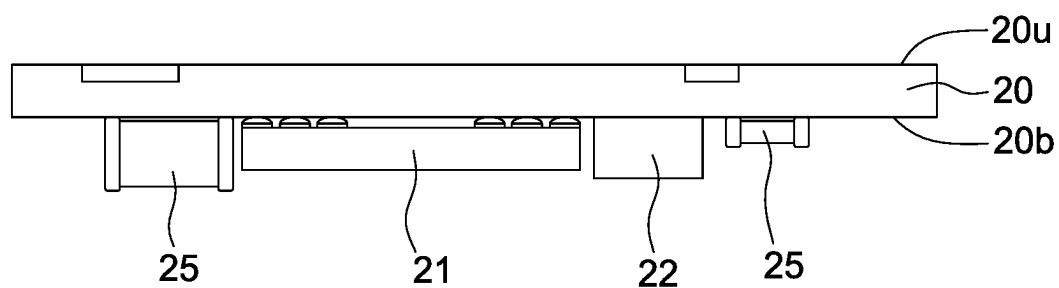


FIG. 2E

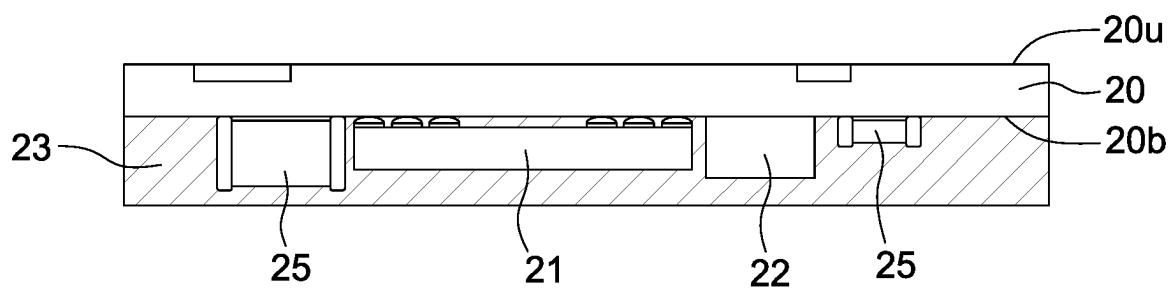


FIG. 2F

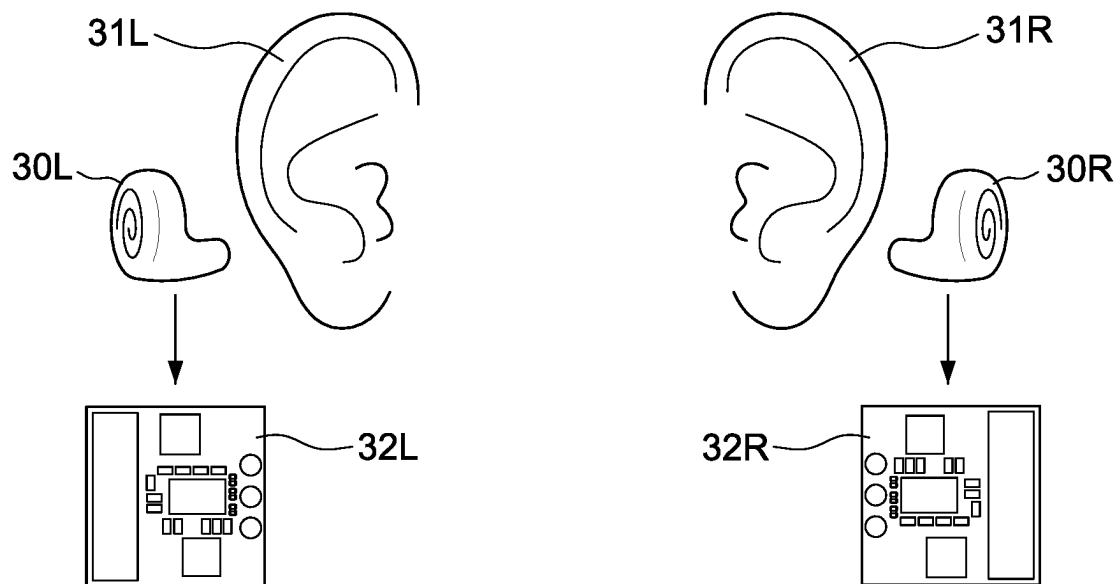


FIG. 3

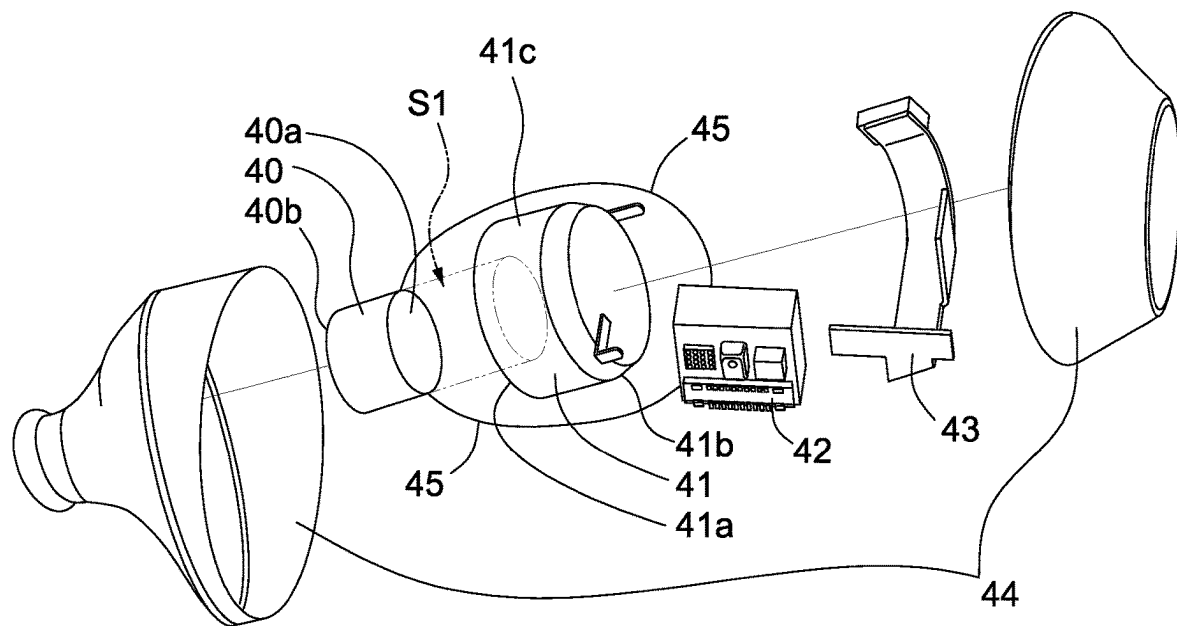


FIG. 4A

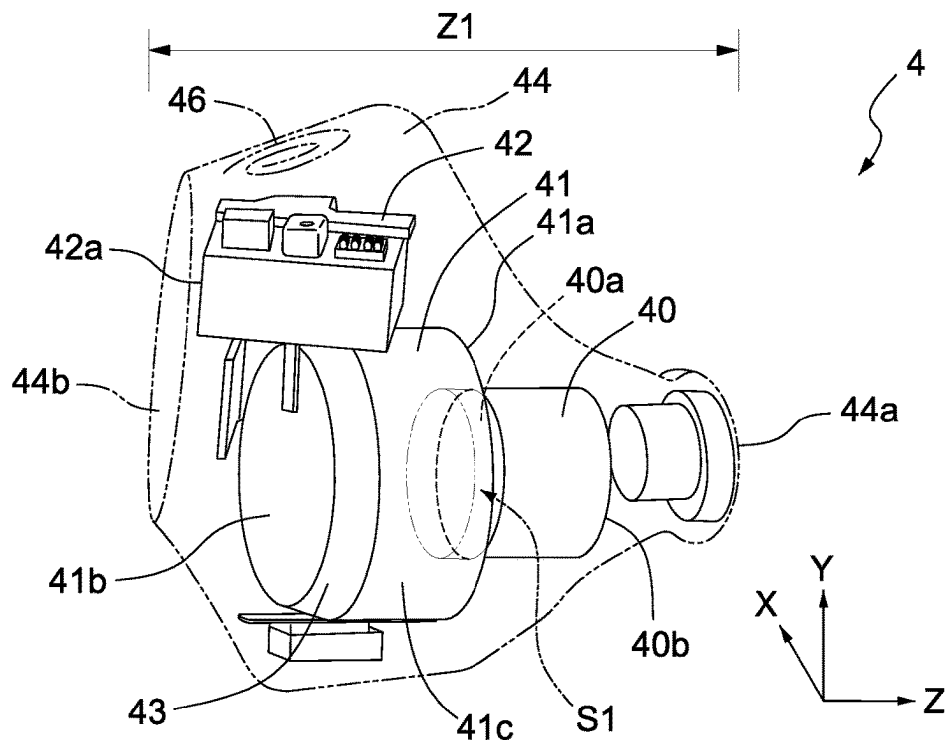


FIG. 4B

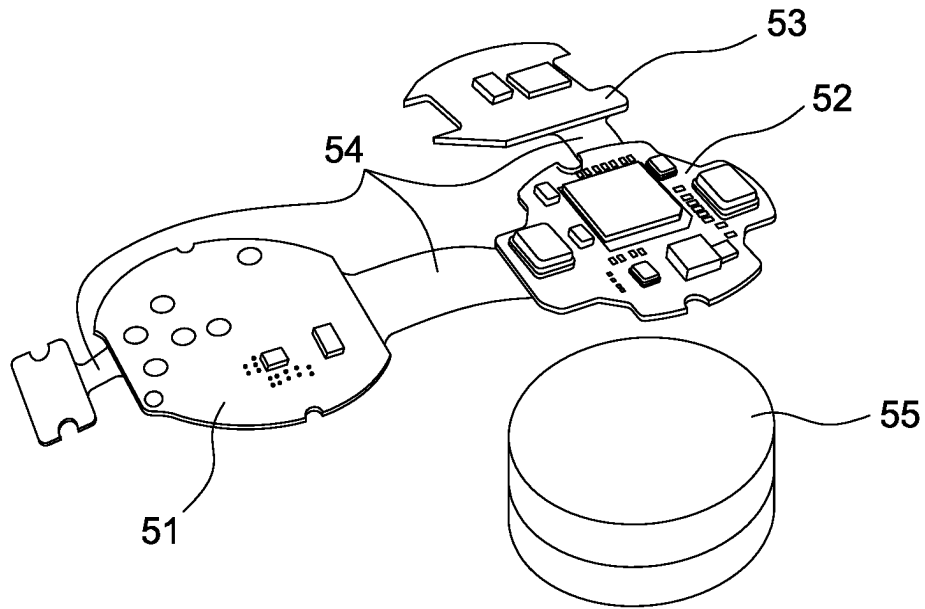


FIG. 5A

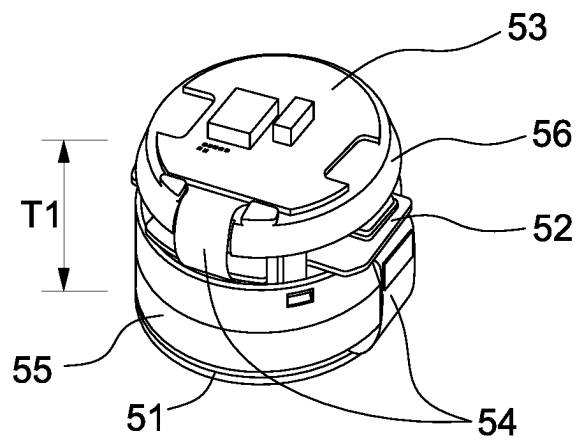


FIG. 5B

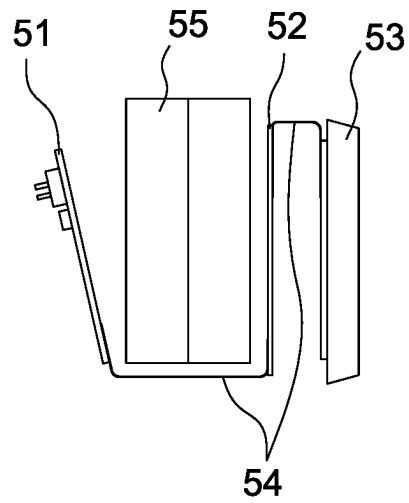


FIG. 5C

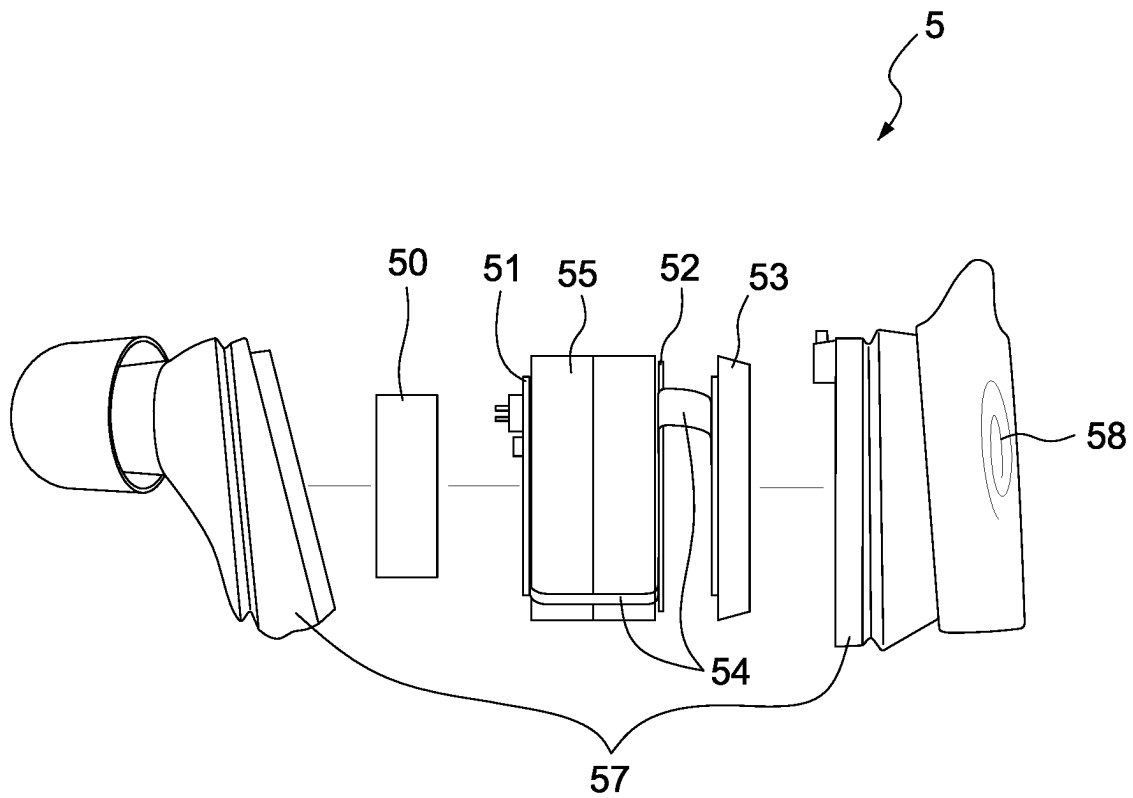


FIG. 5D

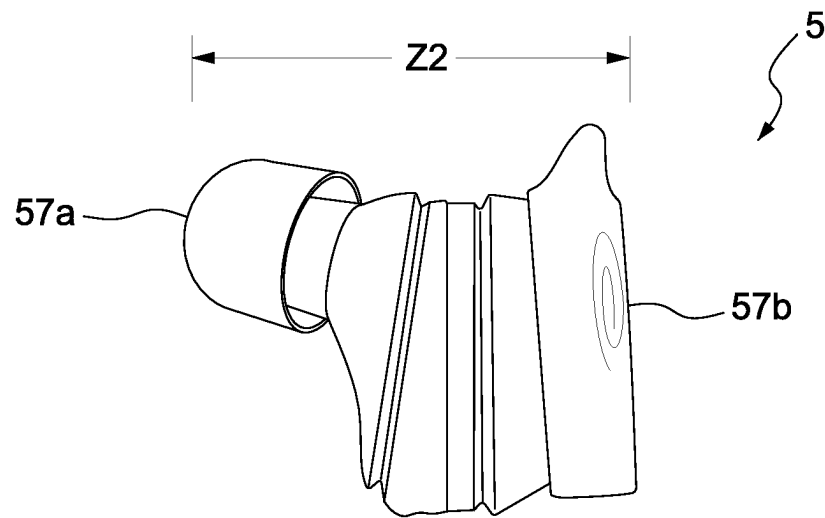


FIG. 5E

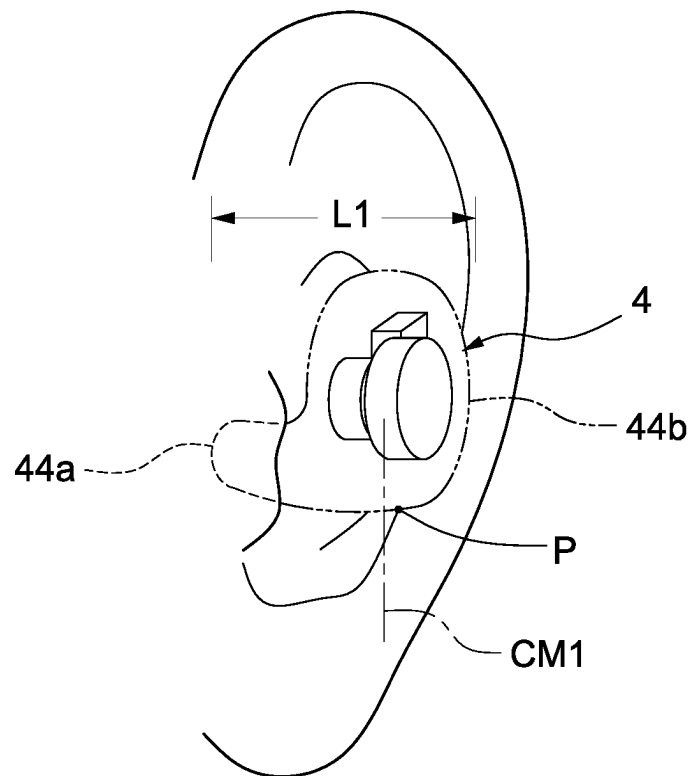


FIG. 6A

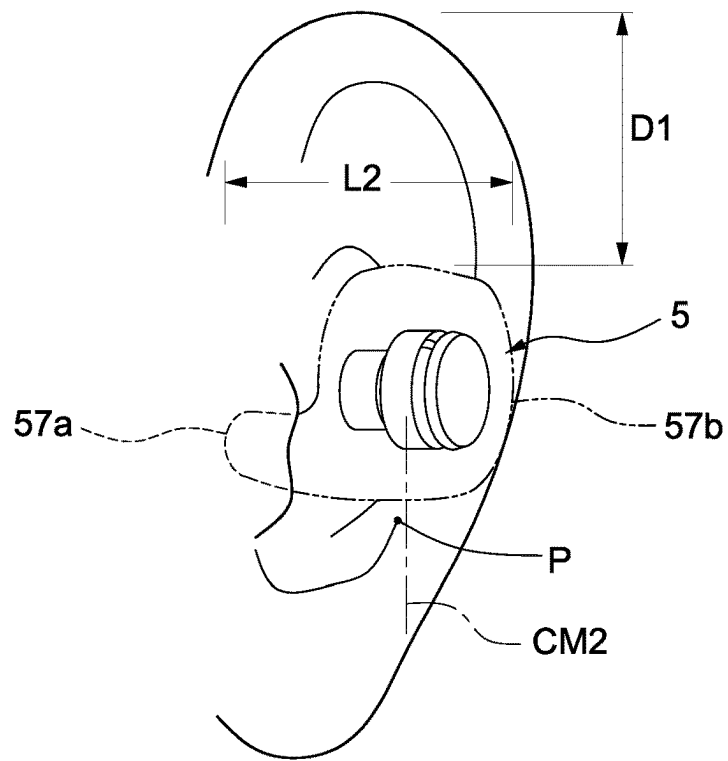


FIG. 6B

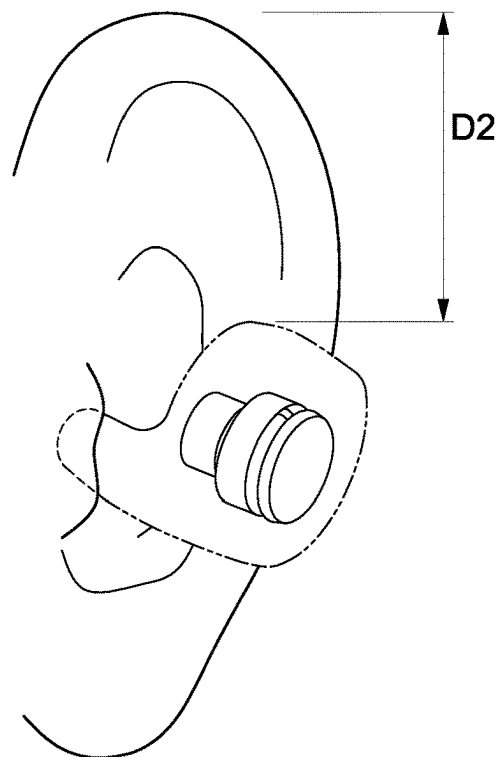


FIG. 6C

1

SEMICONDUCTOR DEVICE PACKAGE AND ACOUSTIC DEVICE HAVING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 16/528,338 filed Jul. 31, 2019, now issued as U.S. Pat. No. 11,477,559, the contents of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates generally to a semiconductor device package, and more particularly, the present disclosure relates to an acoustic device having the semiconductor device package.

2. Description of the Related Art

As technology advances, various circuits or modules can be integrated into one electronic device, e.g. an acoustic device (e.g. an earphone or a wireless earphone) to perform multifunction. However, miniaturization of the electronic device may adversely affect performance of the electronic device.

SUMMARY

In some embodiments, the present disclosure provides a semiconductor device package. The semiconductor device package includes a first substrate, a first electronic component, an encapsulant and a feeding structure. The substrate has a first surface and a second surface opposite the first surface. The first electronic component is disposed on the first surface of the first substrate. The encapsulant encapsulates the first electronic component on the first surface of the first substrate. The feeding structure is disposed on the second surface of the first substrate without covering.

In some embodiments, the present disclosure further provides a wireless earphone. The wireless earphone includes a semiconductor device package. The semiconductor device package includes a first substrate, a first electronic component, an encapsulant and a feeding structure. The substrate has a first surface and a second surface opposite the first surface. The first electronic component is disposed on the first surface of the first substrate. The encapsulant encapsulates the first electronic component on the first surface of the first substrate. The feeding structure is disposed on the second surface of the first substrate without covering.

In some embodiments, the present disclosure provides a semiconductor device package manufacturing method. The method includes providing a substrate having a first surface and a second surface opposite the first surface. The method also includes disposing a first semiconductor device on the first surface of the substrate. The method further includes encapsulating the first semiconductor device on the first surface of the substrate. The method also includes disposing a feeding structure on the second surface of the substrate. The method further includes exposing the feeding structure.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of some embodiments of the present disclosure are readily understood from the following detailed descrip-

2

tion when read with the accompanying figures. It is noted that various structures may not be drawn to scale, and dimensions of the various structures may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates a layout of a semiconductor device package in accordance with some embodiments of the present disclosure.

FIG. 2A illustrates a cross-sectional view of line A-A' of the semiconductor device package shown in FIG. 1.

FIG. 2B illustrates a cross-sectional view of line A-A' of the semiconductor device package shown in FIG. 1.

FIG. 2C illustrates a cross-sectional view of line A-A' of the semiconductor device package shown in FIG. 1.

FIG. 2D, FIG. 2E, and FIG. 2F illustrate a method of manufacturing a semiconductor device package in accordance with some embodiments of the present disclosure.

FIG. 3 illustrates application of the semiconductor device package shown in FIG. 1 in accordance with some embodiments of the present disclosure.

FIG. 4A illustrates an exploded view of an acoustic device in accordance with some embodiments of the present disclosure.

FIG. 4B illustrates a perspective view of an assembly of the acoustic device as shown in FIG. 4A.

FIG. 5A illustrates an exploded view of electronic components of an acoustic device in accordance with some embodiments of the present disclosure.

FIG. 5B illustrates an assembly of the electronic components of the acoustic device in accordance with some embodiments of the present disclosure.

FIG. 5C illustrates a side view of the assembly of the electronic components of the acoustic device in accordance with some embodiments of the present disclosure.

FIG. 5D illustrates an exploded view of an acoustic device in accordance with some embodiments of the present disclosure.

FIG. 5E illustrates a side view of an assembly of the acoustic device as shown in FIG. 5D.

FIG. 6A illustrates an acoustic device in accordance with some embodiments of the present disclosure plugged in a human ear.

FIG. 6B illustrates an acoustic device in accordance with some other embodiments of the present disclosure plugged in a human ear.

FIG. 6C illustrates the acoustic device shown in FIG. 6B falls out of the human ear.

DETAILED DESCRIPTION

FIG. 1 illustrates a layout of a semiconductor device package 1. The semiconductor device package 1 can have a substrate 20, a microphone 10, a controller 11, a sensor 12, a connector 13, a feeding structure 14, ground pads 15 and electronic components 16.

The microphone 10, controller 11, sensor 12, connector 13, feeding structure 14, ground pads 15 and electronic components 16 can be formed on the same side of the semiconductor device package 1.

The microphone 10 is exposed on the substrate 20. The controller 11 is exposed on the substrate 20. The sensor 12 is exposed on the substrate 20. The connector 13 is exposed on the substrate 20. The feeding structure 14 is exposed on the substrate 20. The ground pads 15 are exposed on the substrate 20. The electronic components 16 are exposed on the substrate 20. A conductive pad 17 is disposed on the substrate 20. Conductive pads 18 are disposed on the substrate 20. The conductive pad 17 is exposed on the substrate

20. The conductive pads 18 are exposed on the substrate 20. The conductive pad 17 can be configured or used to connect the battery and the semiconductor device package 1. The conductive pad 18 can be configured or used to connect the speaker and the semiconductor device package 1. The semiconductor device package 1 can include a printed circuit board assembly. Semiconductor devices or semiconductor device packages can be mounted on the semiconductor device package 1. The semiconductor device package 1 can be referred to as a system on module (SOM).

A line A-A' may pass center of the semiconductor device package 1. The line A-A' may pass center of a surface (e.g. top surface, not denoted in FIG. 1) of the substrate 20. The line A-A' may pass the microphone 10 of the substrate 20. The line A-A' may pass the feeding structure 14 of the substrate 20.

A line B-B' may pass center of the semiconductor device package 1. The line B-B' may pass center of the surface of the substrate 20. The line B-B' may pass the microphone 10 of the substrate 20. As shown in FIG. 1, the microphone 10 can be disposed adjacent to the line A-A' of the surface of the substrate 20. The microphone 10 can be disposed adjacent to the line B-B' of the surface of the substrate 20. The feeding structure 14 may be disposed adjacent to the line A-A' of the surface of the substrate 20.

One of the benefits of the layout of the components of the semiconductor device package 1 may include an arrangement of the microphone 10 to be adjacent to or on the center of the top surface (not denoted in FIG. 1) of the substrate 20, which can fit or be used or be assembled in different housings (e.g. a housing for right ear and another housing for left ear).

One of the benefits of the layout of the components of the semiconductor device package 1 may include an arrangement of the feeding structure 14 to be adjacent to or on the center of the top surface (not denoted in FIG. 1) of the substrate 20, which can fit or be used or be assembled in different housings (e.g. a housing for right ear and another housing for left ear).

FIG. 2A illustrates a cross-sectional view across line A-A' as shown in FIG. 1. A semiconductor device package 1a can include a substrate 20, an electronic component 21, another electronic component 22, an encapsulant 23 and a feeding structure 14.

The substrate 20 has a surface 20b and a surface 20u opposite the surface 20b.

The semiconductor device 21 can be disposed on the surface 20b of the substrate 20. The electronic component 22 can be disposed on the surface 20b of the substrate 20. The encapsulant 23 can encapsulate the electronic component 21 on the surface 20b of the substrate 20. The encapsulant 23 can encapsulate the electronic component 22 on the surface 20b of the substrate 20.

The feeding structure 14 can be disposed on the surface 20u of the substrate 20 and is devoid of encapsulant or molding material. The feeding structure 14 is exposed on the surface 20u of the substrate 20. The surface 20u of the substrate 20 is devoid of encapsulant or molding material. Electronic components 25 are disposed on both surfaces 20b and 20u of the substrate 20. The microphone 10 is disposed on the surface 20u. The microphone 10 may include a semiconductor device package. The microphone 10 may include a microelectromechanical systems (MEMS) device. The microphone 10 may include a microelectromechanical systems (MEMS) device package. The microphone 10 is exposed. The sensor 12 is disposed on the surface 20u of the substrate 20. The connector 13 is disposed on the surface

20u of the substrate 20. The surface 20b includes conductive pads for battery connection. The surface 20b includes conductive pads for speaker driver connection. The feeding structure 14 has a surface 14u higher than the electronic microphone 10. The surface 14u is higher than the sensor 12. The surface 14u is higher than the connector 13. The feeding structure 14 is the highest among elements disposed on the surface 20u of the substrate 20.

The electronic components 21 and 22 may include one or more of application processors, memories, digital signal processors, coders/decoders, power management integrated circuits, DC converters, low dropout outputs or filters. The encapsulant 23 may include different kinds of molding materials. The feeding structure 14 includes a spring or a conductive pin. The feeding structure 14 includes a POGO pin. The feeding structure 14 also includes an antioxidation layer. The feeding structure 14 is configured to and can be used to feed a radio frequency (RF) signal to an antenna. The electronic components 25 may include one or more of resistors, inductors or capacitors. The sensor 12 may include a G-sensor or other sensors. The connector 13 includes a zero insertion force connector or a board-to-board connector for peripheral function flexible printed circuit (FPC) connection.

FIG. 2B illustrates another cross-sectional view of line A-A' of the semiconductor device package shown in FIG. 1. Electronic component 21' is disposed on a surface 20'b of a substrate 20'. Electronic component 22' is disposed on the surface 20'b of the substrate 20'. The surface 20'u is opposite to the surface 20'b of the substrate 20'. Electronic components 25 are disposed on both surfaces 20'b and 20'u of the substrate 20'. Encapsulant 23 surrounds the elements disposed on the surface 20'b of the substrate 20'. A substrate 20 has a surface 20b and a surface 20u opposite the surface 20b. A feeding structure 14 is disposed on the surface 20u of the substrate 20. The feeding structure 14 is exposed. The surface 20u of the substrate 20 is devoid of encapsulant or molding material. Electronic components 25 are disposed on both surfaces 20b and 20u of the substrate 20. A microphone 10 is disposed on the surface 20u. The microphone 10 is also exposed. A sensor 12 is disposed on the surface 20u of the substrate 20. A connector 13 is disposed on the surface 20u of the substrate 20. Encapsulant 23 surrounds the elements disposed between the surface 20b of the substrate 20 and the surface 20'u of the substrate 20'.

FIG. 2C illustrates a further cross-sectional view of line A-A' of the semiconductor device package shown in FIG. 1. Electronic component 21" is disposed on a surface 20''b of a substrate 20''. Electronic component 22" is disposed on the surface 20''b of the substrate 20''. The surface 20''u is opposite to the surface 20''b of the substrate 20''. Electronic components 25 are disposed on both surfaces 20''b and 20''u of the substrate 20''. Encapsulant 23 surrounds the elements disposed on the surface 20''b of the substrate 20''. Electronic component 21' is disposed on a surface 20'b of a substrate 20'. Electronic component 22' is disposed on the surface 20'b of the substrate 20'. The surface 20'u is opposite to the surface 20'b of the substrate 20'. Electronic components 25 are disposed on both surfaces 20'b and 20'u of the substrate 20'. Encapsulant 23 surrounds the elements disposed on the surface 20'b of the substrate 20'. A substrate 20 has a surface 20b and a surface 20u opposite the surface 20b. A feeding structure 14 is disposed on the surface 20u of the substrate 20. The feeding structure 14 is exposed. The surface 20u of the substrate 20 is devoid of encapsulant or molding material. Electronic components 25 are disposed on both surfaces 20b and 20u of the substrate 20. A microphone 10 is

5

disposed on the surface **20u**. The microphone **10** is also exposed. A sensor **12** is disposed on the surface **20u** of the substrate **20**. A connector **13** is disposed on the surface **20u** of the substrate **20**. Encapsulant **23** surrounds the elements disposed between the surface **20b** of the substrate **20** and the surface **20'u** of the substrate **20'**. Encapsulant **23** also surrounds the elements disposed between the surface **20'b** of the substrate **20'** and the surface **20''u** of the substrate **20''**.

FIGS. 2D, 2E and 2F illustrates a method for manufacturing the semiconductor device package **1a** shown in FIG. 2A.

As shown in FIG. 2D, a substrate **20** having a surface **20b** and a surface **20u** opposite the surface **20b** is provided.

As shown in FIG. 2E, an electronic component **21** is disposed on the surface **20b** of the substrate **20**. Another electronic component **22** is also disposed on the surface **20b** of the substrate **20**. Electronic components **25** are disposed on both of the surfaces **20b** and **20u** of the substrate **20**. A

As shown in FIG. 2F, the electronic component **21** is encapsulated on the surface **20b** of the substrate **20** by an encapsulant **23**. Another electronic component **22** is encapsulated on the surface **20b** of the substrate **20** by the encapsulant **23**. Electronic components **25** are encapsulated on the surface **20b** of the substrate **20** by the encapsulant **23**. A feeding structure **14** is disposed on the surface **20u** of the substrate **20**. The feeding structure **14** is exposed. A microphone **10** is disposed on the surface **20u** of the substrate **20**. The microphone **10** is exposed. The surface **20u** of the substrate **20** is also exposed. Electronic components **25** are disposed on the surface **20u** of the substrate **20**. A sensor **12** is disposed on the surface **20u** of the substrate **20**. A connector **13** is disposed on the surface **20u** of the substrate **20** to form the semiconductor device package as shown in FIG. 2A.

FIG. 3 illustrates application of semiconductor device packages of earphones. Acoustic device (e.g. earphones) **30R** is plugged into human ears **31R**. Acoustic device **30L** is plugged into human ears **31L**. The right acoustic device **30R** includes semiconductor device packages **32R**. The left acoustic device **30L** includes semiconductor device packages **32L**. The layout of the semiconductor device package **32R** is the same or identical to the layout of the semiconductor device package **32L**, except that the semiconductor device package **32R** is rotated by substantially 180 degrees with respect to the semiconductor device package **32L** after assembly. Accordingly, one layout of semiconductor device package is used for both acoustic devices **30R** and **30L**, which can reduce cost. The acoustic devices **30R** and **30L** may include the semiconductor device packages **1a**, **1b** or **1c** shown in FIGS. 1A, 1B and 1C.

FIG. 4A illustrates an exploded view of an acoustic device **4** in accordance with some embodiments of the present disclosure. The acoustic device **4** includes a speaker **40**, a battery **41**, a semiconductor device package **42**, an interconnect structure **43** and a housing **44**. The speaker has a surface **40a** and a surface **40b** opposite to the surface **40a**. The battery **41** has a surface **41a**, a surface **41b** opposite the surface **41a**, and a surface **41c** extended between the surface **41a** and the surface **41b**. The speaker **40** is disposed adjacent to the surface **41a** of the battery **41**. The surface **40a** of the speaker **40** faces with the surface **41a** of the battery **41**. The structure of the semiconductor device package **42** is illustrated in the cross-sectional view of the semiconductor device package **1a**, **1b** or **1c** in FIGS. 2A-2C, respectively. The speaker **40** is connected to the semiconductor device package **42** by wires **45**. The wires **45** do not penetrate the battery **41**. The speaker **40** is separated from the battery **41**

6

by a space **S1**. There is no turbulent flow in the space **S1**. The space **S1** can influence the acoustic performance of the acoustic device **4**. The housing **44** encloses the speaker **40**, the battery **41**, the semiconductor device package **42**, and the interconnect structure **43**. The interconnect structure **43** connects the battery **41** and the semiconductor device package **42**. The interconnect structure **43** includes flexible print circuit (FPCs) or other interconnect structures.

FIG. 4B illustrates a perspective view of an assembly of the acoustic device **4** as shown in FIG. 4A. The semiconductor device package **42** is disposed adjacent to the surface **41c** of the battery **41**. The semiconductor device package **42** is electrically connected to the surface **41c** of the battery **41** by the interconnect structure **43**. The speaker **40** is separated from the surface **41a** of the battery **41** by the space **S1**. The space **S1** is devoid of electronic components. The space **S1** may have a plate for separating the speaker and the battery **41**. The space **S1** may have conductive wires therein. The space **S1** may have solely conductive wires therein. The space **S1** may have no active components. The space **S1** may have no passive components.

The semiconductor device package **42** is disposed between the housing **44** and the surface **41c** of the battery **41**. The speaker **40** has a surface **40a** facing the surface **41a** of the battery **41**. The semiconductor device package **42** is disposed between the housing **44**, the surface **41c** of the battery **41** and the surface **40a** of the speaker **40**. The semiconductor device package **42** is surrounded by the housing **44**, the surface **41c** of the battery **41** and the surface **40a** of the speaker **40**. The semiconductor device package **42** is disposed in a space defined by the housing **44**, the surface **41c** of the battery **41** and the speaker **40**. The housing **44** encloses the speaker **40**, the battery **41** and the semiconductor device package **42**. The housing **44** has an end **44a** and an end **44b** opposite the end **44a**. Antenna patterns **46** are formed on the housing **44** to work as antennas. The feeding structure **14** may be in contact with the antenna patterns **46** of the housing **44** of the acoustic device **4**.

Referring again to FIG. 4B, assuming the speaker **40** can have a width or diameter of approximately 6 mm, a maximum distance **Z1** between the end **44a** (which can be fitted or plugged into an ear) and the end **44b** along the z axis can be equal to or less than 23 mm. For example, assuming the speaker **40** can have a width or diameter of approximately 6 mm, a distance **Z1** between the end **44a** (which can be fitted or plugged into an ear) and the end **44b** along the z axis can be equal to or less than 20 mm. Accordingly, the semiconductor device package **42** can facilitate miniaturization of the acoustic device **4**. Arrangement of the semiconductor device package **42** can facilitate miniaturization of the acoustic device **4**.

For example, assuming the speaker **40** can have a width or diameter of approximately 10 mm, a distance **Z1** between the end **44a** (which can be fitted or plugged into an ear) and the end **44b** along the z axis can be equal to approximately 23 mm.

FIG. 5A illustrates an exploded view of electronic components of an acoustic device. Circuit boards **51**, **52** and **53** of an acoustic device are connected by an interconnect structure **54**. Electronic components are formed on the circuit boards **51**, **52** and **53**. The interconnect structure **54** may include FPCs or other interconnect structures. A battery **55** can be provided in the acoustic device.

The acoustic device, which includes circuit boards **51**, **52** and **53**, can be fitted or plugged into a left ear. However, another acoustic device to be fitted or plugged into a right

7

ear may include circuit boards each having layout different from circuit boards 51, 52 and 53, which may inevitably increase the cost.

FIG. 5B illustrates an assembly of the electronic components of the acoustic device in accordance with some embodiments of the present disclosure. The circuit boards 51, 52 and 53 are stacked. The thickness of the stack is T1. The circuit boards 51 and 52 and the interconnect structure 54 surround a battery 55. The circuit boards 52 and 53 and the interconnect structure 54 surround a mechanical component 56.

FIG. 5C illustrates a side view of the assembly of the electronic components of the acoustic device. FIG. 5C shows that the circuit board 51 can be above the battery 55. The circuit board 52 can be under the battery 55. The circuit boards 51 and 52 and the interconnect structure 54 surround the battery 55. The circuit board 53 is separated from the battery 55. As shown in FIG. 5C, the mechanical component 56 may be ignored.

FIG. 5D illustrates an exploded view of an acoustic device 5 including the stack shown in FIG. 5C. The acoustic device 5 may include a speaker 50, circuit boards 51, 52 and 53, the interconnect structure 54, a battery 55 and a housing 57. The circuit board 53 is disposed between the speaker 50 and the battery 55, which causes electromagnetic interference to the speaker 50 and thus degrades the acoustic performance of the acoustic device 5. The housing 57 encloses the speaker 50, the circuit boards 51, 52 and 53, the interconnect structures 54, and the battery 55. Antenna patterns 58 are formed on the housing 57. A feeding point (not denoted in FIG. 5D) on the circuit board 53 is in contact with the antenna patterns 58.

FIG. 5E illustrates a side view of an assembly of the acoustic device 5 shown in FIG. 5D. In some embodiments, assuming the diameter of the speaker inside the acoustic device 5 is 6 mm, the maximum distance Z2 between an end 57a (which can be fitted or plugged into an ear) and an end 57b is equal to or larger than 23 mm. Because the thickness T1 shown in FIG. 5B is larger than the distance between an end 42a of the semiconductor device package 42 and the end 41a of the battery 41 along the z axis in FIG. 4B, the maximum distance Z2 of the acoustic device 5 is larger than the maximum distance Z1 of the acoustic device 4.

Referring to FIG. 6A, the acoustic device 4 shown in FIG. 4B is plugged into a human ear. The center of gravity CM1 may be adjacent or close to the end 44b. Due to the miniaturization of the acoustic device 4, when plugging the acoustic device 4 into the ear, the acoustic device 4 can be in contact with a point P of the ear. Accordingly, the acoustic device 4 can have a relatively good support of the ear. Accordingly, the acoustic device 4 would not easily fall out of the even when the person wearing the acoustic device 4 is moving or sweating.

Referring to FIG. 6B, the acoustic device 5 shown in FIG. 5E is plugged into a human ear. The center of gravity CM2 may be adjacent or close to the end 57b.

The acoustic device 5, which may have a relatively great size, may have a major portion (except the plug) that is floating. The acoustic device 5 may be spaced from the point P of the human ear. A distance D1 is between the top surface of the acoustic device 5 and the human ear.

FIG. 6C illustrates that the acoustic device 5 starts falling off the ear when the person wearing the acoustic device 5 swears, moves or change positions. The falling of the acoustic device 5 may be resulted from an unstable state as shown in FIG. 6B, in which the center of gravity CM2 may

8

cause the acoustic device 5, which lacks support of the ear, relatively easy to fall out of the ear.

It is contemplated that a distance D2 from the top surface of the acoustic device 5 and the human ear is increasing due to the falling of the acoustic device 5. The distance D2 is larger than the distance D1.

Spatial descriptions, such as “above,” “below,” “up,” “left,” “right,” “down,” “top,” “bottom,” “vertical,” “horizontal,” “side,” “higher,” “lower,” “upper,” “over,” “under,” and so forth, are indicated with respect to the orientation shown in the figures unless otherwise specified. It should be understood that the spatial descriptions used herein are for purposes of illustration only, and that practical implementations of the structures described herein can be spatially arranged in any orientation or manner, provided that the merits of embodiments of this disclosure are not deviated from by such an arrangement.

As used herein, the terms “approximately,” “substantially,” “substantial” and “about” are used to describe and account for small variations. When used in conjunction with an event or circumstance, the terms can refer to instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, when used in conjunction with a numerical value, the terms can refer to a range of variation less than or equal to $\pm 10\%$ of that numerical value, such as less than or equal to $\pm 5\%$, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$, less than or equal to $\pm 1\%$, less than or equal to $\pm 0.5\%$, less than or equal to $\pm 0.1\%$, or less than or equal to $\pm 0.05\%$. For example, a first numerical value can be deemed to be “substantially” the same or equal to a second numerical value if the first numerical value is within a range of variation of less than or equal to $\pm 10\%$ of the second numerical value, such as less than or equal to $\pm 5\%$, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$, less than or equal to $\pm 1\%$, less than or equal to $\pm 0.5\%$, less than or equal to $\pm 0.1\%$, or less than or equal to $\pm 0.05\%$. For example, “substantially” perpendicular can refer to a range of angular variation relative to 90° that is less than or equal to $\pm 10^\circ$, such as less than or equal to $\pm 5^\circ$, less than or equal to $\pm 4^\circ$, less than or equal to $\pm 3^\circ$, less than or equal to $\pm 2^\circ$, less than or equal to $\pm 1^\circ$, less than or equal to $\pm 0.5^\circ$, less than or equal to $\pm 0.1^\circ$, or less than or equal to $\pm 0.05^\circ$.

Two surfaces can be deemed to be coplanar or substantially coplanar if a displacement between the two surfaces is no greater than $5\text{ }\mu\text{m}$, no greater than $2\text{ }\mu\text{m}$, no greater than $1\text{ }\mu\text{m}$, or no greater than $0.5\text{ }\mu\text{m}$. A surface can be deemed to be substantially flat if a displacement between a highest point and a lowest point of the surface is no greater than $5\text{ }\mu\text{m}$, no greater than $2\text{ }\mu\text{m}$, no greater than $1\text{ }\mu\text{m}$, or no greater than $0.5\text{ }\mu\text{m}$.

As used herein, the singular terms “a,” “an,” and “the” may include plural referents unless the context clearly dictates otherwise.

As used herein, the terms “conductive,” “electrically conductive” and “electrical conductivity” refer to an ability to transport an electric current. Electrically conductive materials typically indicate those materials that exhibit little or no opposition to the flow of an electric current. One measure of electrical conductivity is Siemens per meter (S/m). Typically, an electrically conductive material is one having a conductivity greater than approximately 10^4 S/m , such as at least 10^5 S/m or at least 10^6 S/m . The electrical conductivity of a material can sometimes vary with temperature. Unless

otherwise specified, the electrical conductivity of a material is measured at room temperature.

Additionally, amounts, ratios, and other numerical values are sometimes presented herein in a range format. It is to be understood that such range format is used for convenience and brevity and should be understood flexibly to include numerical values explicitly specified as limits of a range, but also to include all individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly specified.

While the present disclosure has been described and illustrated with reference to specific embodiments thereof, these descriptions and illustrations are not limiting. It should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the present disclosure as defined by the appended claims. The illustrations may not be necessarily drawn to scale. There may be distinctions between the artistic renditions in the present disclosure and the actual apparatus due to manufacturing processes and tolerances. There may be other embodiments of the present disclosure which are not specifically illustrated. The specification and drawings are to be regarded as illustrative rather than restrictive. Modifications may be made to adapt a particular situation, material, composition of matter, method, or process to the objective, spirit and scope of the present disclosure. All such modifications are intended to be within the scope of the claims appended hereto. While the methods disclosed herein have been described with reference to particular operations performed in a particular order, it will be understood that these operations may be combined, sub-divided, or re-ordered to form an equivalent method without departing from the teachings of the present disclosure. Accordingly, unless specifically indicated herein, the order and grouping of the operations are not limitations of the present disclosure.

As used herein, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” “lower,” “left,” “right” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. It should be understood that when an element is referred to as being “connected to” or “coupled to” another element, it may be directly connected to or coupled to the other element, or intervening elements may be present.

The foregoing outlines features of several embodiments and detailed aspects of the present disclosure. The embodiments described in the present disclosure may be readily used as a basis for designing or modifying other processes and structures for carrying out the same or similar purposes and/or achieving the same or similar advantages of the embodiments introduced herein. Such equivalent constructions do not depart from the spirit and scope of the present disclosure, and various changes, substitutions, and alterations may be made without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An electronic device, comprising:
a substrate;
a connector disposed over the substrate;

a feeding structure disposed over the substrate and configured to feed a radio frequency (RF) signal to an antenna; and

a conductive pad disposed adjacent to the connector and configured to connect a speaker, wherein a central axis of a side of a surface of the substrate in a plan view passes the feeding structure.

2. The electronic device of claim 1, wherein the feeding structure is in contact with the antenna.

3. The electronic device of claim 2, further comprising a housing for accommodating the substrate, the connector and the feeding structure, wherein the antenna is formed on the housing.

4. The electronic device of claim 1, wherein the substrate is electrically connected to the speaker through the connector and a flexible printed circuit, wherein the flexible printed circuit connects the connector with the speaker.

5. The electronic device of claim 1, further comprising:
an electronic component disposed under the substrate and electrically connected to the substrate; and
an encapsulant encapsulating the electronic component and in contact with the substrate.

6. The electronic device of claim 1, further comprising a microphone and a sensor disposed between the connector and the feeding structure in the plan view.

7. The electronic device of claim 1, wherein the conductive pad is disposed between the connector and the feeding structure in the plan view.

8. The electronic device of claim 1, wherein the conductive pad is disposed outside a projection of the feeding structure toward the connector in the plan view.

9. An electronic device, comprising:

a housing;

a battery accommodated by the housing; and

a device comprising:

a substrate;

a connector disposed over the substrate;

a conductive pad disposed adjacent to the connector and configured to connect the battery; and

a feeding structure disposed over the substrate and configured to feed a radio frequency (RF) signal to an antenna, wherein a height of the feeding structure is greater than a height of the connector.

10. The electronic device of claim 9, further comprising a flexible printed circuit connecting the connector with the battery.

11. The electronic device of claim 10, wherein a longitudinal direction of the flexible printed circuit is not perpendicular to a longitudinal direction of a lateral surface of the battery.

12. The electronic device of claim 10, wherein a longitudinal direction of the flexible printed circuit is non-parallel to a central axis of the battery.

13. The electronic device of claim 9, wherein the housing includes an antenna pattern, wherein the device is disposed between the antenna pattern and the battery, wherein the device is disposed outside a projection of the battery along an axial direction of the battery.

14. The electronic device of claim 13, wherein the device includes a feeding structure disposed on a first surface of the substrate and in contact with the antenna pattern, wherein the substrate has a second surface opposite to the first surface and facing the battery.

15. The electronic device of claim 9, further comprising an encapsulant encapsulating the substrate, wherein the substrate and the connector are separated from the battery by the encapsulant.

11

16. The electronic device of claim 9, further comprising a speaker accommodated by the housing and separated from the battery by a space, wherein the battery has a first surface facing the speaker and a second surface opposite to the first surface, wherein an extension of the second surface of the battery passes the device. 5

17. An electronic device, comprising:

a housing;

a speaker accommodated by the housing;

a battery accommodated by the housing, wherein the battery has a first surface facing the speaker, a second surface opposite to the first surface, and a lateral surface extending between the first surface and the second surface; 10

a device accommodated by the housing and comprising:

a substrate having a first surface and a second surface opposite to the first surface; and 15

an encapsulant encapsulating the first surface of the substrate; and

a flexible printed circuit disposed over the second surface of the substrate and electrically connecting the sub-

12

strate to the battery and the speaker, wherein the flexible printed circuit is electrically connected to the lateral surface of the battery.

18. The electronic device of claim 17, further comprising a connector disposed on the second surface of the substrate and connecting the substrate with the flexible printed circuit.

19. The electronic device of claim 17, wherein the device further comprises a first electronic component disposed under the first surface of the substrate and encapsulated by the encapsulant, wherein the first electronic component is spaced apart from the battery by the encapsulant.

20. The electronic device of claim 17, wherein the lateral surface of the battery includes a first portion and a second portion opposite to the first portion, wherein a bottom surface of the device faces the first portion of the lateral surface of the battery, wherein the flexible printed circuit extends from a top surface of the device to the second portion of the lateral surface of the battery.

* * * * *